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Lingual articulation of devoiced vowels in Japanese

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01. Introduction

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Phonological rules vs. spontaneous speech processes (phonetics)

- Description of sound patterns in terms of phonological vs. phonetic processes is a **classic debate** in Laboratory Phonology (e.g. Browman and Goldstein, 1990)
- General agreement that some patterns are phonological in nature (i.e., **categorical addition, deletion, change**); others are phonetic (i.e., **gradient overlap** of articulatory gestures) (e.g., Zsiga, 1995, 1997)
- Phonological vs. phonetic nature of a pattern implicate **different models of linguistic knowledge** and, therefore, of **language acquisition and learnability**

Browman, C., & Goldstein, L. (1990). Tiers in Articulatory Phonology: with some implications for casual speech. In Kingston & Beckman (Eds.), *Papers in Laboratory Phonology I: between the grammar and physics of speech* (pp. 341-376). Cambridge: Cambridge University Press.

Vowel devoicing

- Several languages have been reported to **devoice vowels** between voiceless consonants: Greek, Shanghai Chinese, Korean, Montreal French, Tokyo Japanese (Jun et al. 1998)
- These languages have features in common:
 - (1) voicing contrasts in consonants
 - (2) variation in syllable duration (long vs. short syllables).

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Phonetic vs. phonological descriptions of vowel devoicing

- Simple phonetic description:
 - Laryngeal gestures overlap to passively devoice vowels (e.g., Jun and Beckman 1993).

Oral tier:	j	u	t
Laryngeal tier:	devoicing devoicing		

- Simple phonological description:
 - deletion: $V \rightarrow \emptyset$ / [-voice] _____ [-voice] (Kondo, 2001)
 - OR
 - devoicing: $V \rightarrow$ [-voice] / [-voice] _____ [-voice] (Tsuchida, 1997)

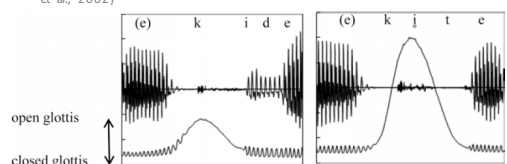
Japanese high vowel devoicing

- Possibly, the best-studied case of vowel devoicing
 - Extensive phonetic work on laryngeal articulation, acoustics of devoicing, and perception of devoiced/deleted vowels (e.g., Hirose, 1971; Beckman and Shoji, 1984; Fujimoto et al., 2002; Whang, 2016) including in child-directed speech (Martin et al., 2015)
 - Phonological arguments for deletion (Kondo, 2001) and opposing arguments for devoicing (Kawahara, 2015)

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Japanese laryngeal gestures

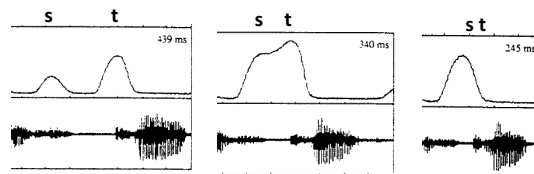
- In devoiced vowel contexts, e.g., /kite/, there is a single laryngeal gesture of **greater magnitude** than a single consonant gesture, c.f. /kide/ (Fujimoto et al., 2002)



Fujimoto, M., Murano, E., Niimi, S., & Kiritani, S. (2002). Differences in glottal opening pattern between Tokyo and Osaka dialect speakers: factors contributing to vowel devoicing. *Folia phoniatrica et logopaedica*, 54(3), 133-143.

“Kiss ted” quickly

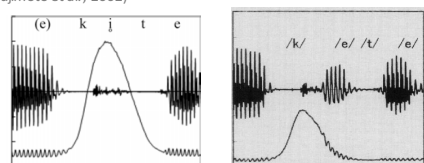
- When laryngeal gestures are brought close together in time, they merge into a single gesture of magnitude equal to sum of component laryngeal gestures (Munhall and Lofqvist, 1992)



Munhall, K., & Lofqvist, A. (1992). GESTURAL AGGREGATION IN SPEECH-LARYNGEAL GESTURES. *Journal of Phonetics*, 20(1), 111-126.

Japanese laryngeal gestures

- The **magnitude** of the laryngeal gesture in Japanese devoicing contexts is **greater than the sum of two voiceless consonant gestures** (Fujimoto et al., 2002)



Fujimoto, M., Murano, E., Niimi, S., & Kiritani, S. (2002). Differences in glottal opening pattern between Tokyo and Osaka dialect speakers: factors contributing to vowel devoicing. *Folia phoniatrica et logopaedica*, 54(3), 133-143.

Summary of past work

- The laryngeal data indicate that devoicing in Japanese is actively controlled.
 - Possibly due to reinterpretation of passive devoicing as active phonological control in modern Tokyo Japanese.
- Acoustic evidence for presence/absence of a lingual vowel articulation has been largely equivocal, with some studies claiming the vowel has been deleted (Beckman and Shoji 1984) while others claiming that it is present (Faber and Vance 2001; cf. Whang 2016)

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Our contribution to the debate

- Electromagnetic Articulography (EMA) data on the **lingual articulation** of vowels in voiced and devoiced contexts.
- A **computational method** for evaluating vowel deletion on the basis of movement kinematics.

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02.

EMA experiment

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EMA experiment

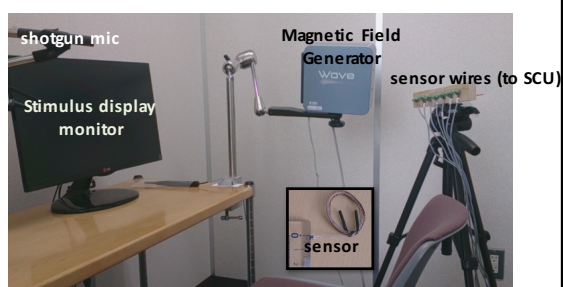
- Investigate lingual articulation of devoiced vowels in environments that vary in predictability.
- Compare articulatory dynamics of devoiced vowels to voiced vowels and to the “deletion trajectory”, generated via simulation.

Materials (target words)

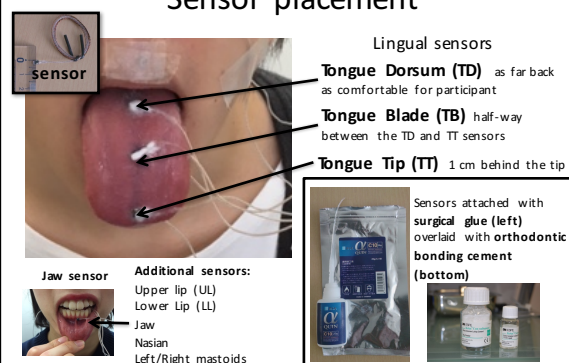
	Devoicing/deletion	Voiced vowel
Entropy = 1.99 (w)	mas <u>u</u> taa マスター	mas <u>u</u> da 益田
Entropy = 1.89 (w)	hak <u>u</u> sai 白菜	yak <u>u</u> zai 薬剤
Entropy = 1.46	j <u>u</u> taisei 主体性	j <u>u</u> daika 主題歌
Entropy = 1.08 (w)	phi <u>u</u> soku 不足	phi <u>u</u> zoku 付属
Entropy = 0.09 (w)	kats <u>u</u> toki 勝つ時	kats <u>u</u> dou 活動

10-15 repetitions of the target words in the carrier phrase: okee _____ to itte 'Ok, say _____'. Participants were instructed to speak as if they were making a request of a friend. Interspersed with 10 words lacking /u/.

Apparatus



Sensor placement



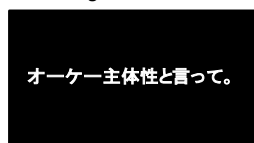
Procedure

- Stimuli were presented on a monitor in random order.
- To promote fluent reading, target words were previewed before displayed in the carrier sentence.
- A native speaker of Japanese monitored pronunciation manually advancing trials after accepting or rejecting each token.

Target preview (500ms)

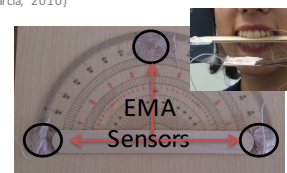
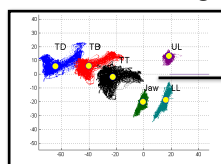


Target sentence



Post-processing

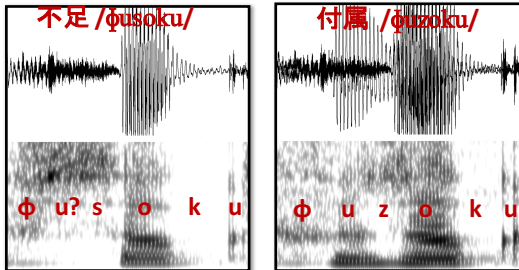
- Head movements corrected computationally
- Data rotated to the occlusal (bite) plane
- Robust smoothing (Garcia, 2010)



Garcia, D. (2010). Robust smoothing of gridded data in one and higher dimensions with missing values. *Computational statistics & data analysis*, 54(4), 1167-1178.

Acoustic results: vowel devoicing

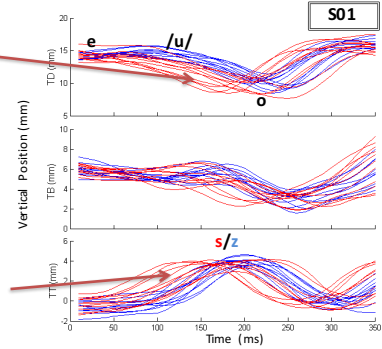
In line with current descriptions of Tokyo Japanese, /u/ was **devoiced** between voiceless consonants and **voiced** otherwise.



Results: fusoku~fuzoku

TD is lower for devoiced vowel

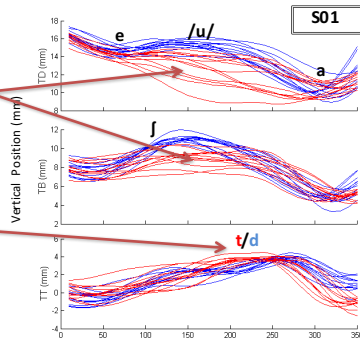
TT rise is earlier for [s] than [z]



Results: shutaisei~shudaika

lower TD and TB for /u/

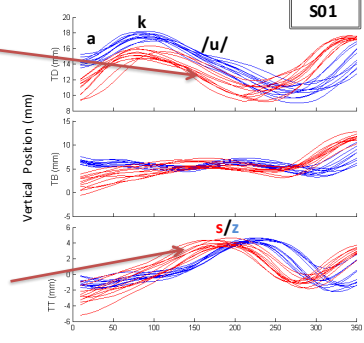
earlier TT gesture for /t/ than for /d/



Results: hokusai~yakuzai

TD is lower for devoiced vowel

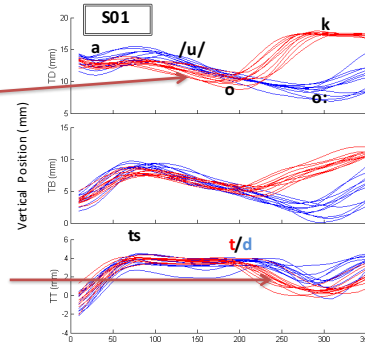
TT rise is earlier after the devoiced vowel



Results: katsutoki~katsudou

TD is lower for devoiced vowel

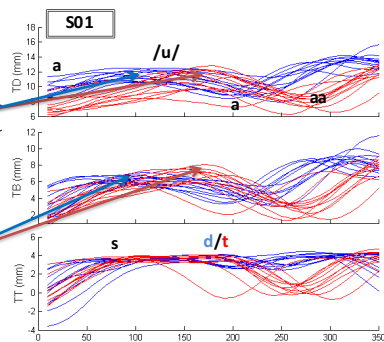
TT release is earlier for the devoiced vowel



Results: masutaa~masuda

Temporal difference in TD rise; devoiced vowel is later

And TB rise; devoiced vowel is later and higher



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03. Phonological deletion vs. phonetic reduction

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Analysis

- **Voicing effect:** do voiced and devoiced vowel trajectories differ?
 - Discrete Cosine Transform (DCT) on TD trajectories
 - MANOVA on DCT coefficients
- **Phonological deletion:** is the /u/ absent in any of these words?
 - Simulate a “targetless” trajectory
 - Compare DCT coefficients of “targetless” trajectory to data, c.f. t-test against zero.

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Discrete Cosine Transform (DCT)

Complex curve represented as the sum of Cosines:

$C(m) \cos(\theta)$
where m is the n th cosine and θ is a function of length

$$C(m) = \frac{2}{N} \sum_{n=0}^{N-1} x(n) \cos\left(\frac{(2n+1)(m-1)\pi}{2N}\right)$$

Where N is the number of data samples;
 $M = 1, \dots, N$; $k_m = \frac{1}{\sqrt{2}}$ when $m = 1$, else $k_m = 1$;
 $x(n)$ is the intercept;

Rao, K. R., & Yip, P. (2014). *Discrete cosine transform: algorithms, advantages, applications*. Academic press.

How many DCT coefficients are needed?

- Nearly lossless compression (99.6%) with 6 coefficients.
- We used 4 DCT coefficients (99.0%)

Interpretation of cosine components

Raw data (green)
Mean DCT (black)
[e]-to-[a] line (red)

1st DCT Coefficient
→ Average TD height

2nd DCT Coefficient
→ V-to-V trajectory

3rd DCT Coefficient
→ Intervening vowel

4th DCT Coefficient
→ Coarticulation

Effect of voicing on lingual articulation

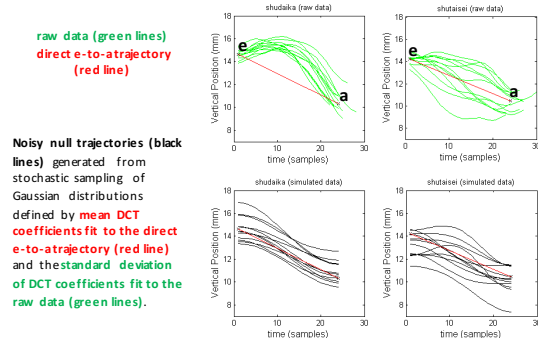
Are the blue (voiced) and red (voiceless) lines significantly different?

To assess the effect of vowel voicing on lingual articulation, we conducted a MANOVA on the DCT coefficients.

S03	Mean and standard deviation of DCT Coefficients			
	1 st Coeff	2 nd Coeff	3rd Coeff	4 th Coeff
shudaika	69.47(3.01)	6.31 (1.59)	-4.54(0.74)	0.94(0.48)
shutaisei	62.18 (6.34)	6.17 (1.83)	-0.04 (2.27)	0.63 (0.95)

$[F(1,23)=23.30, p < .0001***, \text{Wilk's } \Lambda = 0.3209]$

Vowel deletion (“noisy” null) trajectories

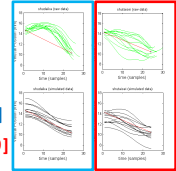


Phonological deletion

To evaluate whether vowels are specified in the phonological output we conducted MANOVAs comparing the DCT coefficients of the data and the “noisy null” hypothesis.

$[F(1,23)= 57.49, p < .0001; \text{Wilk's } \Lambda = 0.0564]$

$[F(1,23)= 3.39, p = .04940; \text{Wilk's } \Lambda = 0.8439]$

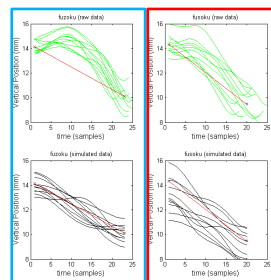


S03	Mean and standard deviation of DCT Coefficients			
	1 st Coeff	2 nd Coeff	3 rd Coeff	4 th Coeff
shudaika (data)	69.47(3.01)	6.31 (1.59)	-4.54(0.74)	0.94(0.48)
shudaika (sim)	60.49(3.01)	5.49(1.59)	-0.00(0.74)	0.61(0.48)
shutaisei (data)	62.18 (6.34)	6.17 (1.83)	-0.04 (2.27)	0.63 (0.95)
shutaisei (sim)	61.21(6.34)	6.30(1.83)	0.00(2.27)	0.70(0.95)

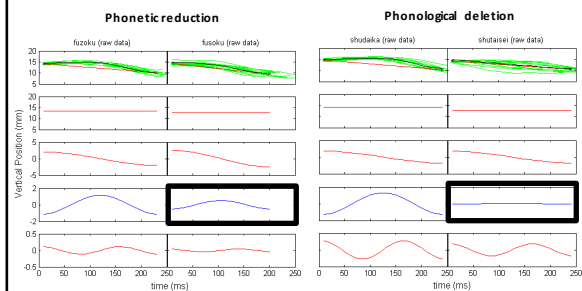
Phonetic reduction in fusoku (c.f. fuzoku)

Main effect of voicing
 $[F(1,21)=22.92, p < .001;$
Wilk's $\Lambda = 0.2798]$

But, both **fusoku** and **fuzoku** are significantly different from noisy null



Phonetic reduction vs. phonological deletion: the view from DCT components



Summary: Vowel reduction, deletion (6 participants)

Target word	Reduction Deletion	
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	(5/6)	(1/6)
f <u>u</u> tasei (c.f., f <u>u</u> daika)	(6/6)	(4/6)
ph <u>u</u> soku (c.f., ph <u>u</u> zoku)	(4/6)	(1/6)
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	(6/6)	(3/6)
mas <u>u</u> taa (c.f., mas <u>u</u> da)	(6/6)	(0/6)
Total: (27/30)		(9/30)

Deletions and reductions pattern together;

Resulting clusters from most to least deletion:
jt >> tst >> fz, kz

Deletion criteria possibly too lax for /aXd/ context as TD rises from /a/ position for /d/

Lingual vowel gestures are reduced/deleted in voiceless contexts

- Vowel height is largely predictable from the large laryngeal gesture devoicing vowels.
- Uninformative gestures are often subject to reduction/deletion (Priva Cohen, 2012,2014; Hume, 2016; Aylett and Turk, 2004, 2006; Bell et al., 2009; Whang, 2016)
- In the absence of the large laryngeal gesture (as for voiceless vowels), the lingual gesture is needed to express vowel height contrast.

Effect of surface consonant cluster type?

- Results are consistent with syllable contact laws (eg, Murray and Venneman, 1983; Venneman, 1987).
- Deletion is most likely when the resulting cluster is fricative stop (FS), followed by affricate stop (aFS), followed by fricative-fricative (FF) and stop-fricative (SF).

FS >> aFS >> FF, SF

Gender differences

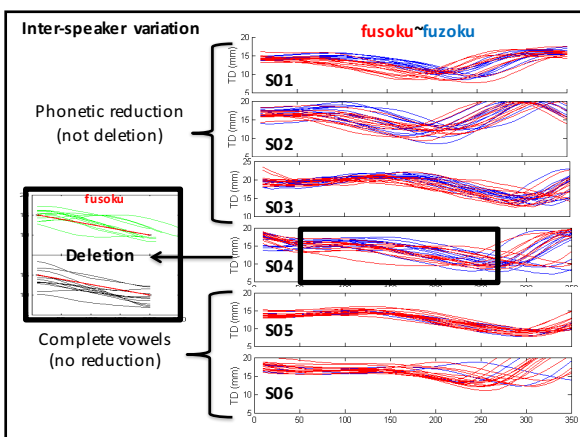
Male female

Target word	Reduction		Deletion	
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	(2/3)	(3/3)	(0/3)	(1/3)
f <u>u</u> taisei (c.f., f <u>u</u> daika)	(3/3)	(3/3)	(2/3)	(2/3)
φ <u>u</u> soku (c.f., φ <u>u</u> zoku)	(2/3)	(2/3)	(1/3)	(0/3)
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	(3/3)	(3/3)	(2/3)	(1/3)
mas <u>u</u> taa (c.f., mas <u>u</u> da)	(3/3)	(3/3)	(0/3)	(0/3)
Totals:		(13/15)	(14/15)	(5/15)

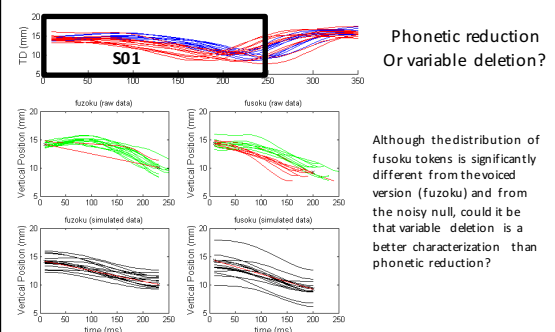
Some limitations of the analysis

- Distributions** of word tokens modelled as **either** deletion **or** reduction
 - this precludes the possibility that individual speakers sometimes delete vowels in a word.
- Noisy null hypothesis is linear, so it doesn't take into account contributions of consonants to TD height (e.g., masutaa)

04. Phonetic reduction as variable deletion



Within-speaker, within-word variation



Token-by-token evaluation

Fit a naïve Bayes classifier to the data and used it to generate (posterior) deletion probabilities

$$p(D|c_1, \dots, c_d) = \frac{p(D) p(c_1, \dots, c_d|D)}{p(c_1, \dots, c_d)}$$

where...

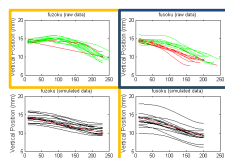
$D = \{\text{deletion, full vowel}\}$

$c_1 = 1^{\text{st}}$ DCT Coefficient

$c_2 = 2^{\text{nd}}$ DCT Coefficient

$c_3 = 3^{\text{rd}}$ DCT Coefficient

$c_4 = 4^{\text{th}}$ DCT Coefficient



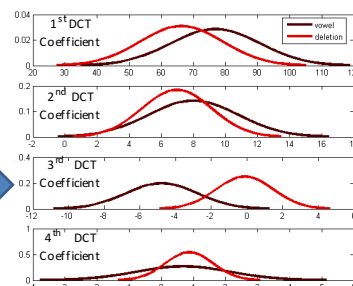
Training data =
voiced tokens &
noisy null

Test data = voiceless
tokens

Classification parameters

/fusoku/

Greatest
separation for
3rd DCT
coefficient



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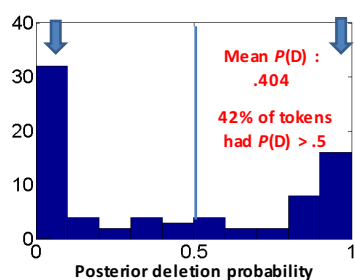
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Deletion probabilities by token - /φusoku/ (all speakers)

Less than .1
chance of deletion

Greater than .9
chance of deletion

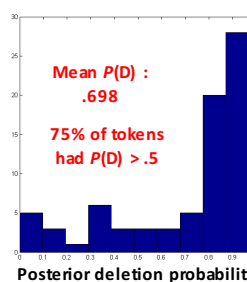
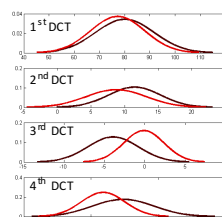


Mean $P(D)$:

.404

42% of tokens
had $P(D) > .5$

Parameters and probabilities /jutaisei/ (all speakers)

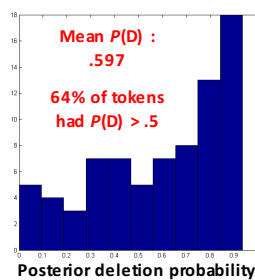
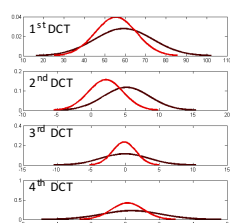


Mean $P(D)$:

.698

75% of tokens
had $P(D) > .5$

Parameters and probabilities /katsutoki/ (all speakers)

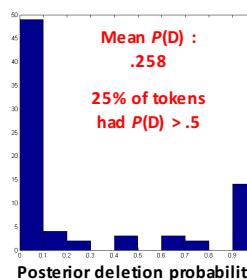
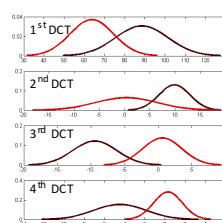


Mean $P(D)$:

.597

64% of tokens
had $P(D) > .5$

Parameters and probabilities /hakusai/ (all speakers)

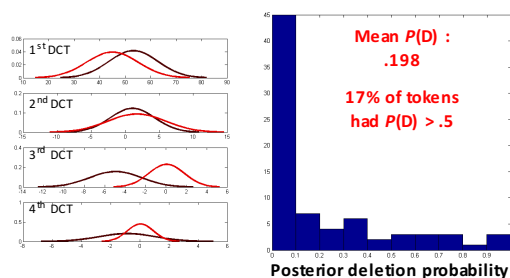


Mean $P(D)$:

.258

25% of tokens
had $P(D) > .5$

Parameters and probabilities /masutaa/ (all speakers)



Summary: deletion probability

Target word	Reduction	Deletion	$P(D)$
<u>hakusai</u> (c.f., yakuzai)	(5/6)	(1/6)	.258
<u>jutaisei</u> (c.f., judaika)	(6/6)	(4/6)	.698
<u>phusoku</u> (c.f., phuzoku)	(4/6)	(1/6)	.405
<u>katsutoki</u> (c.f., katsudou)	(6/6)	(3/6)	.597
<u>masutaa</u> (c.f., masuda)	(6/6)	(0/6)	.198
Total: (27/30)		(9/30)	

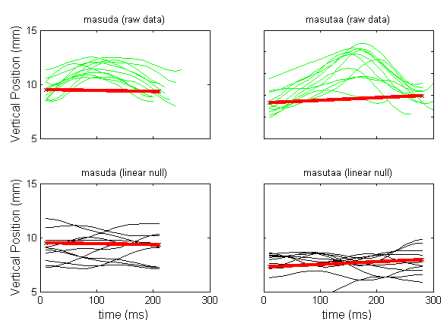
Resulting clusters from most to least deletion:
jt >> tst >> φz >> ks

Token by token analysis

- The token by token analysis indicates that “phonetic reduction” can be understood as variable deletion.
- Vowels are variably deleted in all devoicing contexts, although percentages vary across contexts and speakers.

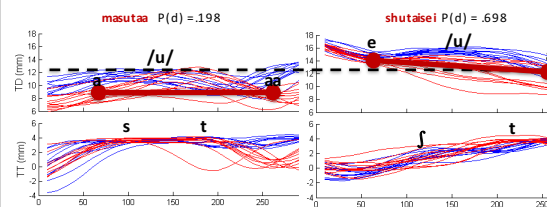
05. A non-linear “noisy” null?

Linear null hypothesis



TD height of /u/ in masutaa

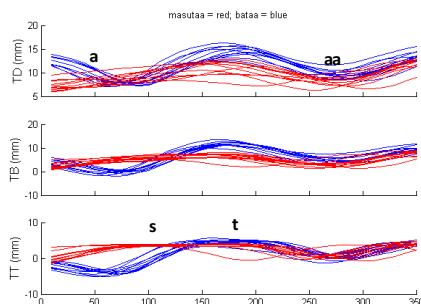
TD is lower in “non-deleted” /u/ in masutaa than in the “deleted” /u/ in shutaisei



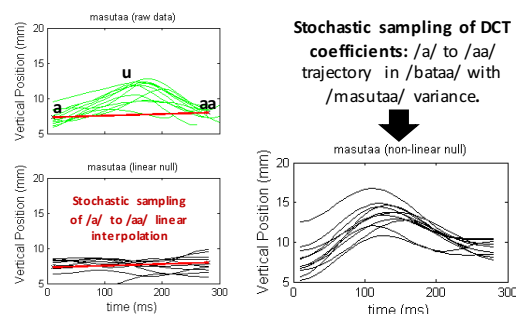
Linear expression of the deletion trajectory may be problematic

TD height: masutaa vs. bataa

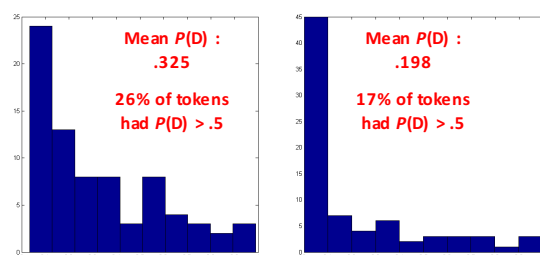
TD raising in /bataa/ is greater than in masutaa and can't be attributed to /u/ in the case of bataa



From linear to non-linear ("noisy" null) trajectories



/u/-deletion based on /bataa/-informed non-linear null



Conclusions: Japanese high vowel devoicing

- Japanese devoiced vowels are characterized by:
 - large laryngeal gestures (past work).
 - And variable deletion of the lingual gesture (this study).
- Deletion probabilities vary across speakers but are uniformly influenced by linguistic factors
 - like other types of well-studied variation (e.g., Guy, 1997; Coetzee and Kawahara, 2013)

More broadly

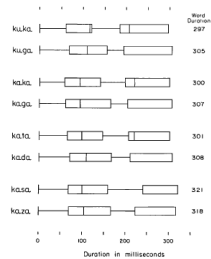
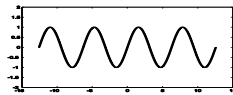
- The structure of phonological form can be rigorously informed by phonetic data.
- We have provided a specific example of a general approach:
 - The key tenet is explicit (and realistic) specification of competing phonological hypothesis in the dimensions of the phonetic data.

04. Acknowledgments

Research supported by the Japan Society for the Promotion of Science postdoctoral fellowship to Jason Shaw and grant-in-aid to Shigeto Kawahara (# 15F15715).

Japanese Mora timing

Japanese segment durations are adjusted to maintain temporal regularity at the mora level (eg, Port et al., 1987)

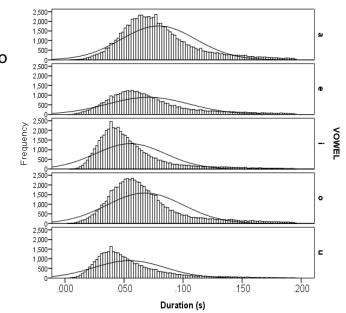


Port, R. E., Dalby, J., & O'Dell, M. (1987). Evidence for mora timing in Japanese. *The Journal of the Acoustical Society of America*, 81(5), 1574-1585.

Japanese vowel duration

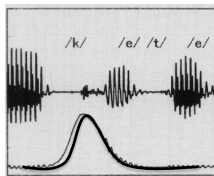
High vowels are also the shortest, leaving less time to implement consecutive devoicing gestures.

	Mean	Median	SD
/a/	79	73	32
/e/	71	63	35
/i/	55	48	30
/o/	68	61	33
/u/	55	47	30



Consecutive voiceless stops

The laryngeal gesture of the second of two voiceless stops is attenuated in magnitude.



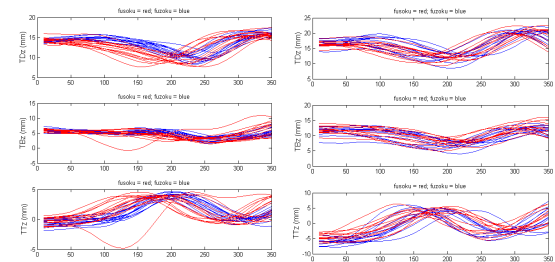
There isn't time for a full laryngeal gesture?

Fujimoto, M., Murano, E., Niimi, S., & Kiritani, S. (2002). Differences in glottal opening pattern between Tokyo and Osaka dialect speakers: factors contributing to vowel devoicing. *Folia phoniatrica et logopaedica*, 54(3), 133-143.

fusoku~fuzoku (two subjects)

S03

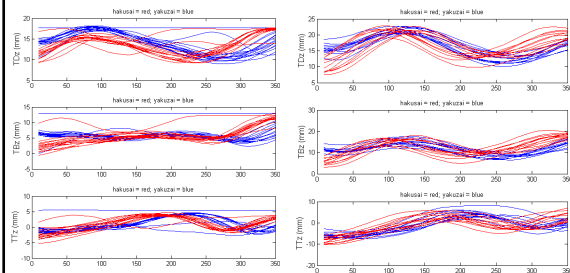
S04



hakusai~yakuzai

S03

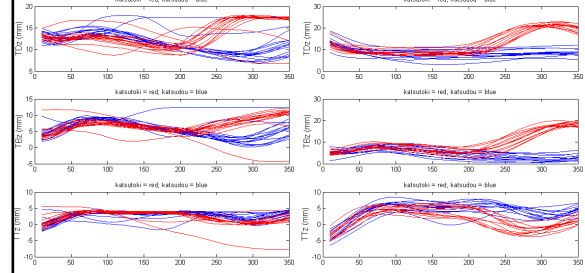
S04



katsutoki~katsudou

S03

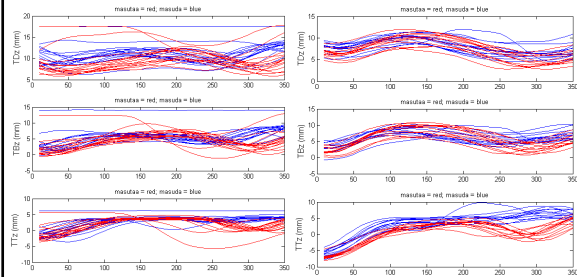
S04



masutaa~masuda

S03

S04



Summary: Reduction or deletion (F3)

Devoicing/deletion	Reduction	Deletion
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	Yes	No
f <u>u</u> taisei (c.f., f <u>u</u> daika)	Yes	Yes
φ <u>u</u> soku (c.f., φ <u>u</u> zoku)	Yes	No
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	Yes	No
mas <u>u</u> taa (c.f., masuda)	Yes	No

Summary: Reduction or deletion (M4)

Devoicing/deletion	Reduction	Deletion
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	No	No
f <u>u</u> taisei (c.f., f <u>u</u> daika)	Yes	No ($p = .03$)
φ <u>u</u> soku (c.f., φ <u>u</u> zoku)	Yes	No
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	Yes	Yes
mas <u>u</u> taa (c.f., masuda)	Yes ($p = .02$)	No

Summary: Reduction or deletion (F5)

Devoicing/deletion	Reduction	Deletion
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	Yes	Yes
f <u>u</u> taisei (c.f., f <u>u</u> daika)	Yes	No
φ <u>u</u> soku (c.f., φ <u>u</u> zoku)	Yes	No
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	Yes	No
mas <u>u</u> taa (c.f., masuda)	Yes	No

Summary: Reduction or deletion (M6)

Devoicing/deletion	Reduction	Deletion
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	Yes	No
f <u>u</u> taisei (c.f., f <u>u</u> daika)	Yes	Yes
φ <u>u</u> soku (c.f., φ <u>u</u> zoku)	Yes	Yes
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	Yes	Yes
mas <u>u</u> taa (c.f., masuda)	Yes	No

Summary: Reduction or deletion (F7)

Devoicing/deletion	Reduction	Deletion
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	Yes	No
f <u>u</u> taisei (c.f., f <u>u</u> daika)	Yes	Yes
φ <u>u</u> soku (c.f., φ <u>u</u> zoku)	No	No
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	Yes	No
mas <u>u</u> taa (c.f., masuda)	Yes	No

Summary: Reduction or deletion (M8)

Devoicing/deletion	Reduction	Deletion
hak <u>u</u> sai (c.f., yak <u>u</u> zai)	Yes	No
f <u>u</u> tai <u>se</u> i (c.f., f <u>u</u> daika)	Yes	Yes
φ <u>u</u> soku (c.f., φ <u>u</u> zoku)	No	No
kats <u>u</u> toki (c.f., kats <u>u</u> dou)	Yes	Yes
mas <u>u</u> taa (c.f., masuda)	Yes	No

Quantifying predictability

- Entropy** is a simple measure of information content from:

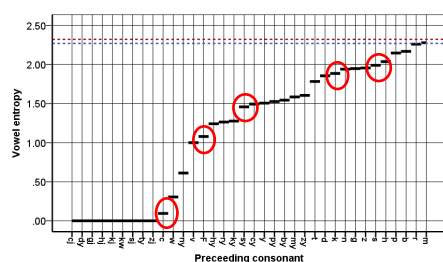
$$H(V) = - \sum_{x \in V} p(x) * \log_2 p(x)$$

- Basic insight: an informative event (high entropy) is one that is not predictable (low probability).

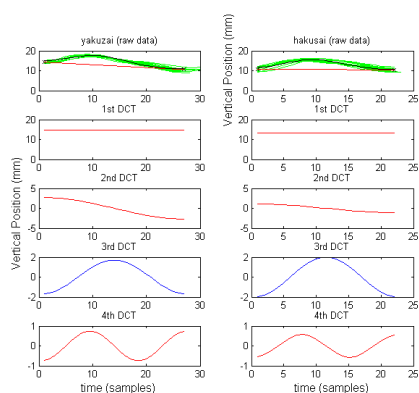
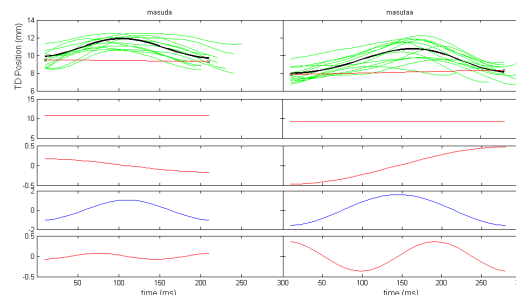
$H(V) = - \sum_{x \in V} p(x) * \log_2 p(x)$ 慶應義塾大学 言語文化研究所 MARCS SCHOOL OF HUMANITIES & COMMUNICATION ARTS

Conditional vowel entropy

- Variation in vowel Entropy as a function of preceding consonant.



DCT components for masuda~masutaa



A non-linear null hypothesis

